

DESIGN INTELLIGENCE

Graphic Design, UI/UX & Long-form Knowledge Base

Principles, decision rules and quality gates for developing a professional design bot

BOT REFERENCE / VERSION 1.0 / SEPTEMBER 2026

Core philosophy

Do not decorate information. Design understanding. The bot should optimize hierarchy, comprehension, usability, accessibility and the intended user action before it optimizes visual novelty.

GRAPHIC

Control attention

UI

Communicate interaction

UX

Reduce effort

LONG-FORM

Sustain understanding

How the design bot should think

This document is both a design primer and a decision framework. The bot should use principles as constraints, not as decorative suggestions.

01 Understand intent

Identify the asset type, audience, objective, content density, platform, dimensions, brand constraints, production format and accessibility requirements.

02 Structure information

Create content hierarchy and semantic groups before choosing visual treatments. Never solve an information problem only with decoration.

03 Select a design system

Choose grid, typography scale, spacing rhythm, color roles, imagery logic and component language. These decisions must be coherent across the full asset.

04 Compose and adapt

Apply the system to each page or screen while preserving continuity. Vary composition intentionally to create rhythm without destroying consistency.

05 Validate

Run usability, accessibility, legibility, consistency and production checks. Correct failures before export.

Bot priority order

1) correctness → 2) comprehension → 3) hierarchy → 4) accessibility → 5) consistency → 6) visual distinction → 7) decoration. Never reverse this order.

Four disciplines, one outcome

Discipline	Primary question	Bot optimization target
Graphic design	What is noticed first, second and third?	Attention and visual hierarchy
UI design	What can be interacted with and what state is it in?	Affordance, feedback and consistency
UX design	Can the user reach the goal easily and confidently?	Task success, reduced friction and clarity
Long-form design	Can the reader stay oriented and engaged across many pages/screens?	Narrative flow, navigation and reading rhythm

The laws of visual hierarchy

The first job of design is to rank information. Every element competes for attention; hierarchy decides who wins and when.

Hierarchy variables

01

Scale

Larger elements generally command attention first.

02

Weight

Boldness increases emphasis but loses power when overused.

03

Position

Top, center, isolation and directional flow change priority.

04

Contrast

Value, color, size and form differences separate levels.

05

Whitespace

Isolation creates perceived importance and reduces cognitive competition.

06

Repetition

Consistent treatments signal equal semantic roles.

Bot rule

Every page/screen must have a clear primary, secondary and tertiary level. If more than three elements compete as primary, the bot must simplify or regroup.

Contrast is relational

Contrast can be created through large/small, bold/regular, dark/light, dense/open, image/text, serif/sans, static/motion, smooth/rough and symmetric/asymmetric relationships. Use contrast to explain importance, not merely to create visual excitement.

Perception before decoration

Humans automatically group visual information. The bot should exploit these grouping tendencies to make structure visible without extra labels.

Principle	What viewers infer	Design-bot application
Proximity	Nearby elements belong together.	Use spacing to create semantic groups before adding boxes.
Similarity	Elements that look alike share a function/category.	Keep components with the same function visually consistent.
Common region	Items inside a boundary form a group.	Use cards/panels only when a real group needs a boundary.
Continuity	The eye follows aligned or directional paths.	Use grids, baselines and directional composition to guide reading.
Closure	Incomplete forms can be perceived as complete.	Use restraint in iconography and brand motifs; do not overdraw.
Figure-ground	Users distinguish focal content from background.	Ensure clear depth, contrast and separation of interactive layers.

Avoid card inflation

Do not put every paragraph inside a card. Cards are for meaningful grouping, actions or independent modules. Excess containers add visual noise and flatten hierarchy.

Proximity test

If the viewer cannot tell what belongs together after removing borders, backgrounds and labels, the spacing system is probably weak. Fix proximity first, then add containers only where necessary.

Build an invisible system

Professional layouts feel coherent because position and spacing are generated from a system rather than guessed element by element.

Grid selection

Context	Recommended starting grid	Why
A4 report / research paper	6 columns	Enough flexibility for text, notes, figures and sidebars.
Editorial / magazine	6 or 12 columns	Supports varied compositions and strong image/text relationships.
Dashboard / desktop web	12 columns	Flexible responsive component placement.
Mobile product UI	4 columns	Simple alignment and predictable responsive behavior.
Presentation / social	Modular grid	Supports visual composition while retaining repeatable anchors.

Spacing system

Use a base spacing unit and scale it systematically. A common digital approach uses 4 and 8 as foundational increments (4, 8, 12, 16, 24, 32, 40, 48, 64...). The exact numbers are less important than consistency and semantic meaning.

Bot rule

The same relationship should receive the same spacing. Example: label-to-field spacing should not randomly be 8 px on one screen, 13 px on another and 18 px on a third.

Golden ratio

The ratio approximately 1:1.618 can be used as a compositional aid for divisions, scales or proportions. It is not a quality certificate. The bot must never force the golden ratio when content, platform constraints or usability suggest a better proportion.

Typography is visible information architecture

A strong type system helps readers understand structure before they consciously read the words.

Semantic type roles

Role	Purpose	Bot behavior
Display	High-emphasis opening statements	Rare; one focal message at a time.
H1	Chapter/page-level title	Strongest structural heading.
H2	Major section	Clearly distinct from body and H3.
H3	Subsection	Supports scanning without overpowering H2.
Lead	Opening/context paragraph	Larger or more relaxed than body.
Body	Sustained reading	Prioritize legibility and rhythm.
Caption	Explain figure/image	Compact but readable; maintain contrast.
Data	Statistics/metrics	Use tabular figures where available; align decimals.
Footnote	Sources/qualifiers	Small but never illegibly small.

Long-form legibility

- Avoid very wide paragraphs. A practical target for sustained reading is often around 50-75 characters per line.
- Line spacing must respond to x-height, font size, line width and weight; wider lines generally need more leading.
- Do not stretch or squeeze fonts. Preserve the typeface's designed proportions.
- Limit typeface count. Use hierarchy through size, weight, case and spacing before introducing another family.

Bot rule

Never label visual hierarchy literally as "Heading", "Subheading" or "Body" in the final design unless the content itself requires those words. Express hierarchy through typography.

Color must communicate, not decorate

A color system should distinguish brand identity, hierarchy, interaction states, semantics and data without compromising accessibility.

Color roles

Role	Examples	Rule
Brand	Primary/secondary palette	Use for identity and emphasis, not every surface.
Neutral	Text, backgrounds, borders	Carry most of the interface/document.
Semantic	Success, warning, error, info	Meaning must also be conveyed by text/icon/shape.
Interaction	Links, focus, selected states	State differences must be perceivable.
Data	Series and categories	Maximize distinction and provide labels/patterns when needed.

WCAG 2.2 minimum gates

Text contrast

Normal text: at least 4.5:1. Large text: at least 3:1. Treat these as minimum thresholds, not aesthetic targets.

Pointer targets

WCAG 2.2 Target Size (Minimum) generally requires at least 24 x 24 CSS px, subject to documented exceptions. Larger targets often improve real usability.

Resize

Web text should support resizing up to 200% without loss of content or functionality under WCAG 2.2.

Accessibility defaults for the bot

- Never rely on color alone to show status or category.
- Maintain visible focus states in interactive products.
- Support logical reading order and semantic headings in digital documents.
- Use alternative text for meaningful images; mark decorative visuals appropriately in supported formats.
- Avoid time-dependent interactions unless users can control them.

Design interaction as a state system

A static screen is only one frame of a product. UI design must communicate what is interactive, what happened and what the user can do next.

Component state completeness

Component	Minimum states to consider
Button	Default, hover (desktop), pressed, focus, loading, disabled, success/error where relevant
Input	Empty, focused, filled, invalid, valid, disabled, read-only, helper/error message
Navigation	Default, hover, active/current, focus, collapsed/expanded
Card/action	Default, hover/select, pressed, loading/empty/error if data-backed
Upload	Idle, drag-over, uploading, processing, success, failed, unsupported file

Nielsen heuristics as bot rules

#	Heuristic	Design-bot rule
01	Visibility of system status	Give timely feedback after actions.
02	Match the real world	Use familiar language and mental models.
03	User control and freedom	Provide undo, cancel and escape where appropriate.
04	Consistency and standards	Use established conventions unless there is a strong reason not to.
05	Error prevention	Prevent likely failures before writing error messages.
06	Recognition rather than recall	Keep needed information visible or easily retrievable.
07	Flexibility and efficiency	Support efficient paths for frequent/expert users.
08	Aesthetic and minimalist design	Remove irrelevant competition from the interface.
09	Recover from errors	Explain the problem in plain language and suggest a fix.
10	Help and documentation	Provide contextual help when the UI cannot be self-evident.

Solve the journey, not the screen

UX begins before visual styling. The bot must reason about the user's goal, context, task sequence, uncertainty and friction.

UX reasoning chain

Required sequence

User → Goal → Context → Evidence → Friction → Information architecture → Flow → Interface → Validation → Iteration

GOV.UK principles worth encoding

01 Start with user needs
Design from researched needs rather than stakeholder assumptions.

02 Do less
Reuse proven patterns and remove non-essential complexity.

03 Design with data
Use behavioral evidence and testing to guide decisions.

04 Do the hard work to make it simple
Hide system complexity instead of passing it to the user.

05 Iterate
Treat design as a learning cycle, not a one-shot output.

06 This is for everyone
Design inclusively from the beginning.

07 Understand context
Consider environment, device, attention, literacy and constraints.

08 Be consistent, not uniform
Preserve coherent patterns without forcing identical layouts everywhere.

Design a reading experience across pages

A long-form document is one continuous information system. It should not feel like a stack of unrelated posters.

Narrative architecture

Macro rhythm

Opening → Orientation → Context → Evidence → Interpretation → Pause → Insight → Application → Resolution

Page rhythm

Avoid repeating the same page recipe throughout a report. Create controlled variation while retaining the same grid, typography, color logic and navigation system.

Page role	Typical visual treatment	Purpose
Opening	Strong title + hero visual / statement	Set expectation and emotional tone.
Orientation	Contents, chapter intro, summary	Tell readers where they are going.
Narrative	Readable text + selective emphasis	Carry the argument.
Evidence	Charts, tables, citations, methodology	Build credibility.
Pause	Quote, image, single statistic	Reset attention and create rhythm.
Synthesis	Framework, diagram, comparison	Convert detail into understanding.
Action	Recommendation, checklist, CTA	Make the next step explicit.

Bot rule

Vary composition, not the design system. A long document should feel alive without feeling inconsistent.

Credibility first, visual clarity second

Research-paper design should improve comprehension and evidence navigation without making the publication feel promotional.

Recommended structure

01	Title & abstract State topic, scope and key takeaway.
02	Research question Define what the study investigates.
03	Context / literature Establish what is already known.
04	Methodology Explain how evidence was gathered.
05	Findings Present observations before interpretation where appropriate.
06	Analysis / discussion Explain meaning, relationships and implications.
07	Limitations Make uncertainty visible.
08	Conclusion Answer the research question and state implications.
09	References / appendices Make evidence traceable and preserve supporting detail.

Research visual rules

- Do not beautify data in ways that obscure scale, uncertainty or comparison.
- Every figure needs a clear purpose, title/caption and source where applicable.
- Keep citation and footnote treatments consistent across all pages.
- Use visual summaries to reveal structure, not to replace necessary methodological detail.
- Use restrained color so evidence remains the focal point.

Show reasoning, not only screens

A credible case study explains why decisions were made and what changed as a result. Final UI images without evidence are portfolio decoration, not a complete case study.

Case-study story arc

Recommended flow

Problem → Context → Evidence → Insight → Design hypothesis → Decision → Prototype → Validation → Outcome → Learning

Translate research into design

Layer	Example	What the bot should make explicit
Observation	Users abandon checkout.	What happened and where.
Evidence	Tests show confusion around account creation.	Source of the finding.
Insight	Users believe registration is mandatory.	Why behavior occurs.
Decision	Introduce guest checkout.	Which design change addresses the insight.
Validation	Completion increases in testing/analytics.	Whether the hypothesis worked.
Learning	Some users still miss shipping cost context.	What remains unresolved / next iteration.

Bot rule

Never invent business impact, conversion uplift, user quotes or research findings. If evidence is absent, label the outcome as a hypothesis, design rationale or expected effect.

Make comparison effortless

The chart type must follow the analytical question, not a desire for novelty.

Question	Prefer	Avoid when possible
Compare categories	Bar / dot plot	3D bars, decorative pictograms
Change over time	Line / area (carefully)	Pie charts across time
Part-to-whole	Stacked bar; pie only for few simple shares	Many-slice donuts
Distribution	Histogram / box plot	Average-only summaries
Relationship	Scatter plot	Dual-axis charts without strong justification
Exact lookup	Table	Over-designed infographic charts

Data integrity gates

- Use zero baselines for bars unless a clearly disclosed analytical reason requires otherwise.
- Do not distort aspect ratios to exaggerate or flatten change.
- Label units, time period, source and sample/context when relevant.
- Use direct labels where they reduce legend-hunting.
- Highlight the insight without hiding the underlying data.

Images are evidence, context or emotion

The bot should know why an image exists. “Because the page looks empty” is not a valid reason.

Image role	Use when	Bot checks
Evidence	Shows a real person, place, product, process or result	Accuracy, permission/source, relevance, caption if needed
Context	Helps the audience understand environment or scale	Authenticity, geography/culture, non-misleading crop
Explanation	Diagram, map, flow or annotated visual reduces complexity	Legibility, labels, semantic order
Emotion	Sets tone or creates a pause in long-form content	Brand alignment, diversity, avoid cliché
Decoration	Adds atmosphere without carrying meaning	Must not compete with content or reduce accessibility

Image quality gate

Never stretch images. Preserve aspect ratio. Avoid pixelated logos, low-resolution exports, accidental cropping of faces/text, and AI-generated imagery that contradicts factual content.

Visual consistency

A long-form asset should define an image treatment system: photography style, crop behavior, corner treatment, illustration style, icon stroke/weight and chart aesthetic. Variation is allowed within the system, not outside it.

Adapt hierarchy, not just dimensions

A design that is merely scaled down is not responsive. The bot should recompose content according to viewport, reading distance and interaction mode.

When space decreases	Preferred response
Text becomes too narrow	Reduce columns; preserve readable line length.
Navigation no longer fits	Change navigation pattern; do not shrink labels into illegibility.
Data table overflows	Prioritize columns, allow horizontal scroll or provide alternate summary.
Hero content dominates mobile	Reduce decorative scale and elevate task-critical content.
Multiple cards become dense	Stack, group or progressively disclose; avoid tiny grids.
Desktop hover affordances disappear	Provide touch-visible cues and states.

Bot rule

Preserve semantic priority across formats. The desktop layout may be 3 columns and mobile 1 column, but the most important information should remain the most important.

Convert principles into repeatable decisions

A capable design bot should not select layouts randomly. It should infer constraints and select patterns based on content structure and user goals.

Required inputs

Input class	Examples
Asset	Research paper, case study, e-book, annual report, website, dashboard, social carousel
Audience	CXO, consumer, government official, technical user, internal team, investor
Objective	Educate, persuade, explain, compare, convert, inform, document
Content	Word count, sections, tables, charts, images, citations, attachments
Brand	Logo, colors, fonts, voice, templates, image style, mandatory placements
Output	A4 PDF, 16:9 slides, web responsive, 1080x1350, editable source
Constraints	Accessibility, legal copy, production specs, localization, deadline

Decision rules

01 Content-heavy → editorial system
Prioritize reading width, typography, navigation, page rhythm and evidence presentation.

02 Task-heavy → interaction system
Prioritize flow, state completeness, feedback, error prevention and action clarity.

03 Data-heavy → analytical system
Prioritize comparisons, annotation, source context and visual integrity.

04 Brand-heavy → identity system
Use brand assets consistently but preserve accessibility and content hierarchy.

05 Multi-format → responsive system
Define semantic priority first, then adapt composition per format.

06 Uncertain content → conservative output
Do not invent facts, citations, metrics, people or product capabilities.

Design from source content without corrupting meaning

When the bot receives PDF, DOC/DOCX, PPT/PPTX, XLS/XLSX or CSV input, redesign must preserve the source structure and factual content before improving presentation.

Ingestion priorities

01 Extract structure
Identify titles, headings, paragraphs, lists, tables, figures, charts, captions, notes and page/slide order.

02 Preserve meaning
Do not silently rewrite factual claims, numbers, units, names, citations or legal text.

03 Map content roles
Classify every content block by semantic role before styling it.

04 Rebuild hierarchy
Improve information architecture when safe, while retaining traceability to source content.

05 Redesign visuals
Apply grid, typography, spacing, brand and image treatment appropriate to the requested output.

06 Validate completeness
Compare redesigned output against source to ensure no content or data was unintentionally dropped.

Non-negotiable

Source text, tables, charts, images, slide/page order, headings and structure should be correctly interpreted. If extraction confidence is low, the bot should flag uncertainty rather than hallucinate missing content.

A design is not done when it exports

The bot should score every output and automatically revise failures before delivering the asset.

Category	Weight	Pass criteria
Content fidelity	20%	No invented or missing critical content; numbers and names verified.
Hierarchy	15%	Primary/secondary/tertiary priorities are obvious.
Legibility	15%	Readable type, line length, spacing and image quality.
Consistency	10%	Grid, typography, components, spacing and brand rules repeat coherently.
Accessibility	15%	Contrast, targets, focus/state cues and non-color meaning pass relevant gates.
UX/task clarity	10%	Actions, states, navigation and errors are understandable.
Visual craft	10%	Balanced composition, image treatment and production polish.
Export quality	5%	Correct dimensions, no clipping, broken glyphs, stretching or pixelation.

Recommended delivery threshold

Do not deliver below 85/100. Any failure in content fidelity, accessibility-critical issues, unreadable text, clipped content or broken interaction states is a blocking defect regardless of total score.

Pre-export checklist

- Correct dimensions and orientation
- No clipped/overlapping elements
- No stretched text or images
- Logos are sharp and correctly proportioned
- Heading levels and reading order are coherent
- All interactive elements have necessary states
- Contrast/accessibility checks pass
- Numbers, sources and legal copy match the source
- Repeated elements align to the same grid
- Final file format matches the request

A reusable instruction layer for the bot

The following rules can be embedded into the design agent's system prompt or quality-control layer.

Primary instruction

Think like a senior graphic designer, editorial designer and product designer. Treat content structure, user goal, accessibility and production accuracy as constraints. Use design principles to make decisions; do not apply decorative templates blindly.

Design-generation rules

- Analyze all supplied content before selecting a layout.
- Generate a content hierarchy and semantic structure before visual styling.
- Use a consistent grid, spacing scale, typography hierarchy, color roles and visual language.
- Create variation through composition while preserving system consistency across long-form pages.
- For UI, design complete component states and give immediate feedback for consequential actions.
- For UX, minimize memory burden, prevent errors, support recovery and keep next steps visible.
- For research/case studies, distinguish evidence, insight, design rationale and outcome.
- For data, choose charts based on the comparison question and never visually manipulate the evidence.
- For file-based design, preserve source facts, structure, tables, figures and sequence unless explicitly asked to edit content.
- Validate output using the QC rubric before export; revise any blocking defect.

Anti-patterns the bot must reject

- Same template for every output
- Random gradients/cards merely to fill space
- Too many font sizes or typefaces
- Everything bold / everything centered
- Tiny body copy to fit excess content
- Pixelated logos
- Unlabeled icons for critical actions
- Color-only error/success indicators
- Decorative charts that obscure values
- Repeated page layouts with no long-form rhythm
- Invented research claims or results

The compact version

Use this page as the final runtime checklist for every design generation request.

#	Check	Question
01	Purpose	What must the design achieve?
02	Audience	Who needs to understand or use it?
03	Content integrity	What must not change?
04	Hierarchy	What is first, second and third?
05	Grid	What invisible structure controls alignment?
06	Typography	Is sustained reading effortless?
07	Spacing	Do distances communicate relationships?
08	Contrast	Is importance and interaction perceivable?
09	Consistency	Do repeated things behave and look repeated?
10	Accessibility	Can more people perceive and operate it?
11	Interaction	Does every action have feedback and recovery?
12	Rhythm	Does long-form content vary without losing continuity?
13	Evidence	Are charts, citations and claims traceable?
14	Production	Are size, resolution, crop and export technically correct?
15	Validation	Would the intended audience understand it without a designer explaining it?

Final principle

A beautiful report that nobody can read has failed. A beautiful app that confuses people has failed. A beautiful dashboard that hides the important number has failed. Design succeeds when it produces the intended understanding or behavior with the least unnecessary cognitive effort.

Reference foundation

These sources anchor the bot's usability and accessibility rules. Product-specific guidance should be re-verified as standards evolve.

[1] Nielsen Norman Group 10 Usability Heuristics for User Interface Design

<https://www.nngroup.com/articles/ten-usability-heuristics/>
Last reviewed Jan 30, 2024.

[2] W3C / WAI Web Content Accessibility Guidelines (WCAG) 2.2

<https://www.w3.org/TR/WCAG22/>
W3C Recommendation; includes contrast, resizing and interaction criteria.

[3] W3C / WAI Understanding SC 1.4.3: Contrast (Minimum)

<https://www.w3.org/WAI/WCAG22/Understanding/contrast-minimum.html>
Defines 4.5:1 normal text and 3:1 large text minimums at AA.

[4] W3C / WAI Understanding SC 2.5.8: Target Size (Minimum)

<https://www.w3.org/WAI/WCAG22/Understanding/target-size-minimum>
Defines 24 x 24 CSS px minimum target size subject to exceptions.

[5] Apple Human Interface Guidelines - Design Principles

<https://developer.apple.com/design/human-interface-guidelines/design-principles>
Current Apple design principles, including purpose, flexibility, simplicity and craft.

[6] Apple Human Interface Guidelines - Accessibility

<https://developer.apple.com/design/human-interface-guidelines/accessibility>
Accessibility guidance for perceivable, adaptable and intuitive interfaces.

[7] GOV.UK Government Design Principles

<https://www.gov.uk/guidance/government-design-principles>
Start with user needs; design with data; simplify; iterate; design for everyone.

Maintenance note

Re-validate accessibility standards, platform HIGs and browser/device requirements periodically. Principles are durable; implementation details evolve.

END OF DESIGN BOT KNOWLEDGE BASE / V1.0